Ross McIlvaine

Full Stack Software Engineer

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Highly adaptable, motivated, and organized software developer currently pursuing a Master's in Computer Science. My experience includes a 4-year degree in Computer Engineering and 3+ years of professional experience in full-stack development. My industry development experience demonstrated my ability to quickly learn and apply industry-standard development practices. I'm seeking a role where I can continue learning and improving my skills while making a meaningful impact on my team and the world around me.

Education

University of Illinois Urbana-Champaign

Master of Computer Science — Chicago, IL

Planned Graduation Date: May 2027

University of Notre Dame

Bachelor of Science in Computer Engineering — Notre Dame, IN

GPA: 3.8, Dean's List (4), Graduated cum laude — Graduation Date: May 2022

Experience

Chamberlain Group | AI Software Engineering Team — Oak Brook, IL

AI Development Intern — May 2026 – August 2026

- Fine-tuned OSNet re-identification model, achieving 58% mAP and 51% Rank-1 on a deliberately hard evaluation set
- Built a ground-truth dataset of 900 videos spanning 600 unique vehicle IDs using company video data
- Designed and implemented the evaluation pipeline benchmarking mAP and Rank-1 retrieval metrics
- Engineered a gallery maintenance algorithm that dynamically updates the vehicle database as new detections occur; experiments projected 85–90% accuracy
- Optimized model inference for cost efficiency alongside retrieval accuracy

Capstone Experience | Drug Interaction Database — Chicago, IL

Database Engineer | *Team Member* — January 2026 – May 2026

- Researched clinical drug-interaction challenges and gathered data sources to define the problem space
- Designed a Neo4j graph schema and authored Cypher queries to model complex drug relationships
- Led UI design for the final product, enabling non-technical users to explore potential adverse drug reactions

Epic Systems | Data Warehouse and Analysis Application Team — Verona, WI

Full-Stack Developer — August 2022 – August 2024

- Spearheaded new functionality to create foreign key joins between external datasets and Epic's data model
- Lead UI and API Developer for a data integration application serving over 5,000 users
- Conducted SQL performance and code review for development projects across the company

Projects

Personal Portfolio Website — Spring 2024

- Built web app from scratch using Svelte framework along with TypeScript and Tailwind CSS
- Developed components to be reactive, modular, and visually appealing
- Deployed to the web using Cloudflare Pages

Skills

Languages/Frameworks: C/C++, C#/.NET, SQL, Python, HTML, CSS, TypeScript/JavaScript, React, Java, Svelte

Computer: Git Version Control, Linux, Amazon Web Services, Microsoft Office